MARKING SCHEME OF COMPUTER SCIENCE

FORM TWO MOCK EXAMINATION-2026

ARUSHA [redacted]

SECTION A(15%marks)

QN 1.

QUESTIONS	i	ii	Iii	iv	V	vi	vii	viii	ix	x
ANSWERS	A	B	B	B	B	D	B	B	B	D

1@ 1marks x 10 = 10 marks

QN2.

QUESTIONS	i.	ii.	iii.	iv.	v.
ANSWERS	E	D	B	F	C

1@ marks x 5 = 5marks

SECTION B(70%Marks)

QN3.

a) The first-generation computers were large, over-heated, and power inefficient, because: they used vacuum tubes for processing, vacuum consumed a lot of power, produced much heat and were bulky in size. 2marks

b) The development of microprocessors was a major development in computer technology, because microprocessors combined many computer functions into a single chip, making computers smellers, faster, cheaper and more efficient. 2marks

c) Mobile phone is regarded as computer because:

- It has input devices such as touch screen and keypad
- It has storage capacity for saving data and applications
- It has outputs features such as screen display and sound
- Can connected to internet and other devices just like computer
- It has processing ability to executes programs.

Only 3 points @ 1 marks= 3marks

d) Central processing unit (CPU), sometimes called brain of computer because:

- It executes instructions
- It processing data
- It controls all operations
- It makes decision
- It manages system performance

- Offers flexible learning
- Enable access to educations without need physical classroom
- Reduces transportations cost
- Supports blended learning (combine online and in- person methods)
- Offers online materials or resources at time when required
- Keeps learning updated with time as resources or contents to online regularly updated

Only 5 points @ 1marks = 5 marks

b. Key concepts and ideas that forms foundation of computer science are:
   - Algorithms. This is step-by step set of instructions used to solve problem
   - Data structures. Organizing and storin data for example arrays
   - Programming. Writing instructions or code that computer can understand and executes, uses language like java, c, c++ and python.
   - Problem solving. Ability to analyze problem and design solutions logically
   - Computer software and hardware. Involves both programs or instructions that run on hardware and physical parts of computer.

   Only 5points @ 1 marks = 5marks

QN5.

A. Computer care and handling; Refers to the proper way of using, maintaining and protecting a computer to ensure it works efficiently, lasts longer and avoids damage. 2marks
B. The effects of dust inside computer are:
   - Reduced cooling efficient: As block air vents cooling fans and heat sinks within computer
   - Increased noise: After cooling fans affected with dust, they must work hard to maintain temperature often result raise up fan noise
   - System instability and crashes: This happens due to system overheating as result of dust accumulations
   - Shorten component life spans: Excessive heat due to dust cause stress on internal components
   - Impaired performance: reduce the system performance within the computer.
   - Fire hazard.

Only 5 points @ 1marks = 4marks

C. Importance of using cc and Bcc when using e-mail appropriately;

- Maintain privacy of recipient's e-mail address
- Keeps relevant people informed
- Prevent confusion and unnecessary replies from many recipients

Only 3 points @ 1 marks = 3marks

QN6

a) Compiler is a program that translates entire source code of a programs into a machine code before executions **while** interpreter is a program that translates and executes the source code line by line at time run. Example python, Ruby, JavaScript .(4marks)

b) Factors that John must consider before start installing software to computer:
   - System requirements; check if your computer has enough CPU power, RAM, and disk space
   - Compatibility; make sure software works with your operating system and existing hardware.
   - Source and safety; Only download from trusted sources to avoid viruses and malware
   - License and cost; known whether it is free or paid before start installing
   - Storage space; Ensure there is enough hard disk space for installation
   - Dependencies; Some software needs other programs or libraries example java

Only 6 points @ 1 marks = 6 marks

QN7. a) Pseudocode; is a simple, informed way of writing steps of a program using plain language instead of a specific programming language. (2marks)

b) *Solution*

START (1mark)
INPUT Fahrenheit (2mark)
celsius = $\frac{5}{9}$ x (Fahrenheit - 32). (3marks)
DISPLAY Celsius (2marks)
END. (1mark)

QN8. a) Trouble shooting; is the process of finding and fixing problems in a system, device or process. (2marks)

b) Four (4) aspects of trouble shooting are:
   - problem identification

- root cause analysis
- isolating and testing
- solution implementation
- verification
- documentation

only 4points @ 2marks = 8marks

QN9.

a) Best practices of cloud security are:
   i. Use of strong password
   ii. Keep software updated
   iii. Limit access. Apply principle of least privilege which give users only access when needs.
   iv. Backup data
   v. Shared responsibility. Users must be secured their data while providers handle the infrastructure.

Only 4 points @ 1 marks = (4marks)

b) Computer hardware; Refers to all the physical parts of a computer that you can see and touch. 1 mark
   Five categories of computer hardware are:
   i. Input devices; example keyboard, mouse, scanner.
   ii. Output devices; examples monitor, printer and speakers
   iii. Processing devices; example CPU
   iv. Storage devices; examples hard disk, flash drive
   v. Communication devices; used to send and receive data between computers. examples modem, network card, router

Only 5 points @ 1mark = 5 marks

SECTION C(15%Marks)

10 ⟶ introductions

- Definition of authentication (2 marks)

  ⟶ Main body

- Methods of authentication; which are:
  - Password verification; The system checks whether the users enter the correct password that matches the stored credentials
  - Username and account identification; system verifies the unique username or account number associated with bank or company account
  - Two-factor authentication(2FA): After entering login details the system may send verification code to the user's mobile phone or e-mail for confirmation
  - Biometric authentication; The system uses fingerprint, face recognition or other biological features that confirm identity
  - Security questions verification; The system may ask for security questions that only the authorized user can answer correctly.
  - Device or location recognitions; The system may check device information or monitor login location to detect unusual access.

Only 6 points @ 2marks = (12 marks)

⟶ **Conclusion (any related) (1 marks)** **COMPUTER SCIENCE**